

Homework 3: The Ionosphere

Answer Key

Fall 2025

1 Plotting the Ionosphere

Question Use the [International Reference Ionosphere \(IRI-2012\)](#) hosted by the CCMC. Run the model to create a text file changing only the following parameters:

Variable	Value
Year, Month, Day	2006, Dec. 21
Time	Local
Hour of day	Noon
Latitude, Longitude	40 N, -105 E (255 W)
Step size	5 km
IRI Model Parameters	e^- , O^+ , H^+ , O_2^+ NO^+ and He^+

(i) Plot altitude (on the y-axis) vs. number density (log scale, on the x-axis) for electrons, O^+ , H^+ , O_2^+ , NO^+ , and He^+ . Use your plot to answer the following questions:

(ii) Below 500 km, in what altitude range does NO^+ dominate? At what altitude does O_2^+ contribute more than 20% of the total?

(iii) What is the total electron content (TEC)? What is the TEC in the topside ionosphere (region above the peak electron density)? What is the TEC in the bottom side ionosphere (region below the peak electron density)?

Answer (i) The following Python code is an example of how to plot the IRI data, as a reference for you. Your plot should look like the one this code produces.

```
1 import pandas as pd
2 from matplotlib import pyplot as plt
3 plt.style.use('seaborn')
4 import seaborn as sns
5
6 # Load data
7 col_names = ['H', 'Ne', 'NeNmF2', 'Tn', 'Ti', 'Te', 'O+', 'N+', 'H+', 'He+', 'O2+', 'NO+', 'Clust', 'TEC', 't']
8 iri = pd.read_csv('iri_output.txt', delim_whitespace=True, skiprows=30, names=col_names, na_values=[-1,0])
9
10 # Get everything into m^-3
11 iri['Ne'] = iri['Ne']*1e6 #m^-3
12 iri['O+'] = 1e-2*iri['O+']*iri['Ne'] #m^-3
13 iri['H+'] = 1e-2*iri['H+']*iri['Ne'] #m^-3
```

```

14 iri['O2+'] = 1e-2*iri['O2+']*iri['Ne'] #m^-3
15 iri['NO+'] = 1e-2*iri['NO+']*iri['Ne'] #m^-3
16 iri['He+'] = 1e-2*iri['He+']*iri['Ne'] #m^-3
17
18 # Plot
19 colpal = sns.color_palette("hls", 10)
20 plt.figure(figsize=(5,6), dpi=300)
21 plt.semilogx(iri['Ne'], iri['H'], label='Ne', c=colpal[4])
22 plt.semilogx(iri['O+'], iri['H'], label='O+', c=colpal[5])
23 plt.semilogx(iri['H+'], iri['H'], label='H+', c=colpal[6])
24 plt.semilogx(iri['O2+'], iri['H'], label='O2+', c=colpal[7])
25 plt.semilogx(iri['NO+'], iri['H'], label='NO+', c=colpal[8])
26 plt.semilogx(iri['He+'], iri['H'], label='He+', c=colpal[9])
27 plt.legend()
28 plt.xlabel(r'Number Density [m$^{-3}$]')
29 plt.ylabel('Altitude [km]')
30 plt.show()

```

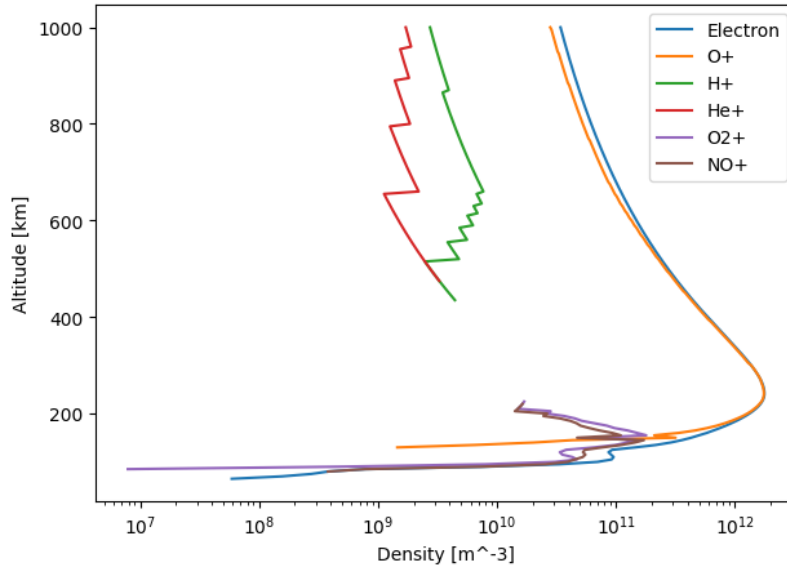


Figure 1: Altitude vs. number density plot for Problem 2(i).

(ii) Below 500 km, NO^+ dominates from about 80 to 145 km. O_2^+ contributes more than 20% of the total density between 95 and 145 km.

(iii) TEC is the integral of electron density in a 1 m^2 column:

```

1 import pandas as pd
2 import numpy as np
3
4 # Load data
5 col_names = ['H', 'Ne', 'NeNmF2', 'Tn', 'Ti', 'Te', 'O+', 'N+', 'H+', 'He+', 'O2+', 'NO+', 'Clust', 'TEC', 't']
6 iri = pd.read_csv('iri_output.txt', delim_whitespace=True, skiprows=30, names=col_names, na_values=[-1])

```

```

7
8 # Convert units
9 iri['Ne'] = iri['Ne']*1e6 #m^-3
10
11 # Calculate TEC
12 TEC = np.trapz(iri[~np.isnan(iri['Ne'])]['Ne'], iri[~np.isnan(iri['Ne'])]['H']*1e3)

```

The TEC from the whole file is 39.6×10^{16} electrons per m^2 or 39.6 TECU, with a peak electron density around 295 km. The topside TEC is about 27.1 TECU and the bottomside TEC is about 11.6 TECU. The topside and bottomside TEC will vary slightly depending on which part you included the peak electron density in (or if you included it in both or neither).

2 Peak Ionization

Question In a Chapman layer, the peak ionization production rate, $q_{\max}$ is related to the electron density $n_{e,\max}$ and the effective recombination coefficient α_{eff} by $n_{e,\max} = [q_{\max}/\alpha_{\text{eff}}]^{1/2}$.

(i) Using the following parameters, determine $q_{\max}$ for each layer of the ionosphere:

Region	Peak Altitude	$n_{e,\max}$ (cm^{-3})	α_{eff} (cm^3/s)
E	110 km	1.5×10^5	10^{-8}
F1	200 km	2.5×10^5	7×10^{-9}
F2	300 km	5.0×10^5	5×10^{-9}

(ii) Explain why, when all other factors are kept constant, the ionization peak ($z_{\max}$) of a layer moves upward with increasing solar zenith angle χ .

Answer (i) All we need to do is solve the given equation for $q_{\max}$ and plug in the values for each region!

$$n_{e,\max} = [q_{\max}/\alpha_{\text{eff}}]^{1/2} \rightarrow q_{\max} = n_{e,\max}^2 \alpha_{\text{eff}}$$

$$q_{\max,E} = (1.5 \times 10^5 \text{ cm}^{-3})^2 \times (10^{-8} \text{ cm}^3/\text{s}) \simeq 225 \times 10^6 \text{ m}^{-3}\text{s}^{-1}$$

$$q_{\max,F1} = (2.5 \times 10^5 \text{ cm}^{-3})^2 \times (7 \times 10^{-9} \text{ cm}^3/\text{s}) \simeq 438 \times 10^6 \text{ m}^{-3}\text{s}^{-1}$$

$$q_{\max,F2} = (5.0 \times 10^5 \text{ cm}^{-3})^2 \times (5 \times 10^{-9} \text{ cm}^3/\text{s}) \simeq 1250 \times 10^6 \text{ m}^{-3}\text{s}^{-1}$$

(ii) The zenith angle χ is zero at local noon when the Sun is directly overhead. χ increases as the sun gets closer to the horizon. As χ increases, the solar radiation propagates through a longer, slant path through the atmosphere compared to $\chi = 0$. It will deposit its energy (and create ionization) based on the total amount of atmosphere traversed; this amount of atmosphere will be reached sooner in altitude for $\chi > 0$, and so the energy will be deposited higher in altitude.

3 Spacecraft Charging in LEO

Question What is the floating potential of an unbiased 6U CubeSat in low-Earth orbit at 400 km altitude, flying with its 20×30 cm face in the ram direction? Assume an electron temperature of 1000 K. What assumptions did you have to make?

Answer Use Eq. 23 from the Ionosphere notes (eq. 4.30 from Tribble),

$$V_{fl} = \frac{kT_e}{q_e} \ln\left(\frac{4v_0 A_i}{v_{e,th} A_e}\right)$$

where k is Boltzmann's constant, T_e is the electron temperature, q_e is the elementary charge, v_0 is the orbital speed, A_i is the area of the area of the spacecraft that is collecting ions (ram side), $v_{e,th}$ is the thermal speed for the electrons, and A_e is the area collecting electrons (the whole surface).

$$v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{(6.674 \times 10^{-11} \text{ N kg}^{-2} \text{ m}^2)(5.972 \times 10^{24} \text{ kg})}{400 \times 10^3 \text{ m} + R_E}} = 7,670 \text{ m/s}$$

$$v_{e,th} = (1.6 \pm 0.2) \sqrt{\frac{kT}{m}} = (1.6 \pm 0.2) \sqrt{\frac{(1.381 \times 10^{-23} \text{ J K}^{-1})(1000 \text{ K})}{(9.11 \times 10^{-31} \text{ kg})}} = 172,000 \text{ to } 222,000 \text{ m/s}$$

$$A_i = 20 \text{ cm} \times 30 \text{ cm} = 0.06 \text{ m}^2$$

$$A_e = 2(20 \times 30) \text{ cm}^2 + 2(10 \times 20) \text{ cm}^2 + 2(10 \times 30) \text{ cm}^2 = 0.22 \text{ m}^2$$

$$V_{fl} = \frac{kT_e}{q_e} \ln\left(\frac{4v_0 A_i}{v_{e,th} A_e}\right) = \boxed{-0.283 \text{ V to } -0.261 \text{ V}}$$

The major assumption here is that this equation is valid for a 6U CubeSat! The equation was derived for a spherical spacecraft, and is not strictly correct for other shapes. However it is a reasonably good first approximation.

4 Spacecraft Charging in GEO

Question Consider a spacecraft in GEO with one surface in sunlight (normal to the sun's direction) and one surface in shadow. Use equation (1) and the relevant equations that follow to calculate the floating potential on each of these two surfaces. Assume a plasma density of $n_e = n_i = 0.5 \text{ cm}^{-3}$, and a plasma temperature $T_e = T_i = 5 \times 10^6 \text{ K}$. You can ignore the SEE and backscatter currents. What is the potential difference between these two surfaces?

$$\frac{dQ}{dt} = I_e(V) + I_i(V) + I_{SEE_i}(V) + I_b(V) + I_{ph}(V) \quad (1)$$

[Hint: you can assume the potential is negative in shadow and positive in sunlight. Note that you'll probably need to solve the equation for dQ/dt numerically.]

Answer The goal is to find at what potential $dQ/dt = 0$ for each side. First, generate a range of potentials. Next, calculate I_e , I_i , and I_{ph} from the potentials. Finally, add them together to calculate dQ/dt , and see at what potential dQ/dt is closest to zero. repeat this process for the other side. An example Python script for this problem is shown below:

```
1 import numpy as np
2 from matplotlib import pyplot as plt
3 plt.style.use('seaborn')
4 import seaborn as sns
5
```

```

6 # Floating potential in Geostationary orbit
7
8 # physical constants
9 qe = 1.6e-19
10 me = 9.11e-31
11 mi = 1.67e-27
12 kB = 1.38e-23
13 kBTph = 2*qe
14
15 # environment parameters
16 ne = 0.5e6
17 ni = 0.5e6
18 Te = 5e6
19 Ti = Te
20
21 A = 1
22 theta = 0
23
24 Iph0 = 40e-6
25
26 veth = np.sqrt(3*kB*Te/(me))
27 vith = np.sqrt(3*kB*Ti/(mi))
28
29 V1 = np.arange(-2000, 0, 1)
30 V2 = np.arange(1, 2000, 1)
31
32 ## calculate currents
33
34 # for negative voltages:
35
36 Ie1 = -0.25 * ne * qe * veth * A * np.exp(qe*V1/(kB*Te))
37 Ii1 = 0.25 * ni * qe * vith * A * (1 - qe*V1/(kB*Te))
38 Iph1 = 0
39
40 # for positive voltages:
41
42 Ie2 = -0.25 * ne * qe * veth * A * (1 + qe*V2/(kB*Te))
43 Ii2 = 0.25 * ni * qe * vith * A * np.exp(-qe*V2/(kB*Ti))
44 Iph2 = Iph0 * A * np.cos(theta) * np.exp(-qe*V2/(kBTph))
45
46 # add them together and find where dQ/dt = 0
47
48 dQdt1 = Ie1 + Ii1 + Iph1
49 dQdt2 = Ie2 + Ii2 + Iph2
50
51 idx1 = (np.abs(dQdt1)).argmin()
52 print(V1[idx1])
53
54 idx2 = (np.abs(dQdt2)).argmin()
55 print(V2[idx2])
56
57 # plot
58
59 fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15,5), dpi=300)
60
61 ax1.plot(V1,dQdt1)
62 ax1.scatter(V1[idx1], dQdt1[idx1], c='m', marker='*', lw=2, zorder=2)
63 ax1.set_xlabel('V')
64 ax1.set_ylabel('dQ/dt')
65 ax1.set_title('Negative Potential')
66
67 ax2.semilogx(V2,dQdt2)
68 ax2.scatter(V2[idx2], dQdt2[idx2], c='m', marker='*', lw=2, zorder=2)
69 ax2.set_xlabel('V')

```

```

70 ax2.set_ylabel('dQ/dt')
71 ax2.set_title('Positive Potential')
72
73 plt.show()

```

We find $dQ/dt \simeq 0$ on the negative side at -1080 V and on the positive side at 10 V. Therefore, the potential difference is $|-1080 - 10| = \boxed{1090 \text{ V}}$. Figure 2 shows dQ/dt vs. V for each side, with the points closest to $dQ/dt \simeq 0$ starred. **Comment:** the actual electron density in geostationary orbit is much lower (2-3 orders of magnitude) than what was assumed here. Thus, the negative potential on the shadow side would be much lower than the answer shown for this problem.

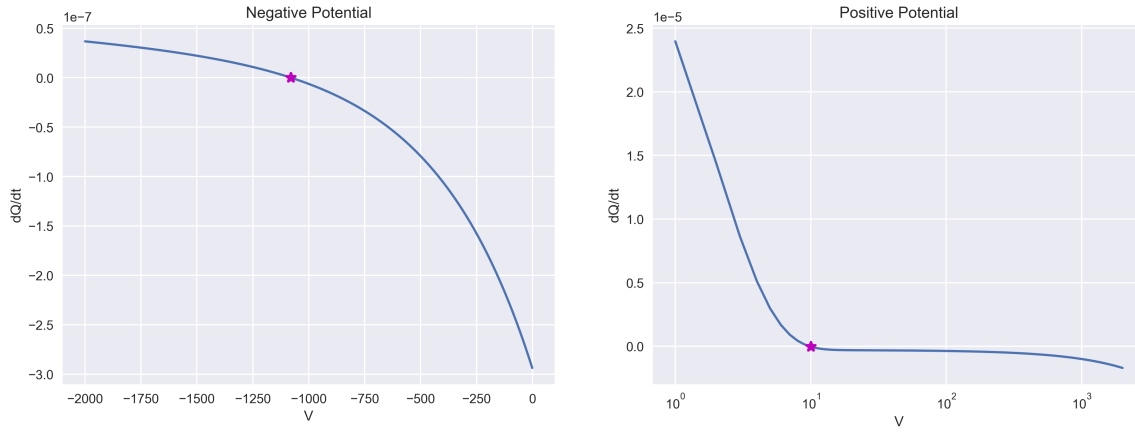


Figure 2: dQ/dt vs. V for Problem 4.

5 Radio Wave Propagation

Question Using the IRI profile you created in Problem 1:

- (i) What is the minimum radio frequency that can be used to communicate from ground to space?
- (ii) From which ionospheric region does this frequency reflect?
- (iii) From what altitude does a 2 MHz radio wave reflect?

Answer (i) The critical frequency, the highest frequency that will be reflected by the ionosphere, is given by:

$$f_c = 9 \times 10^{-3} \sqrt{n_{e,max}}$$

where $n_{e,max}$ is the maximum electron density in cm^{-3} and f_c is the critical frequency in MHz. From our IRI profile, $n_{e,max} = 1,770,444 \text{ cm}^{-3}$. Our critical frequency is then:

$$f_c = 9 \times 10^{-3} \sqrt{1,770,444} = \boxed{12 \text{ MHz}}$$

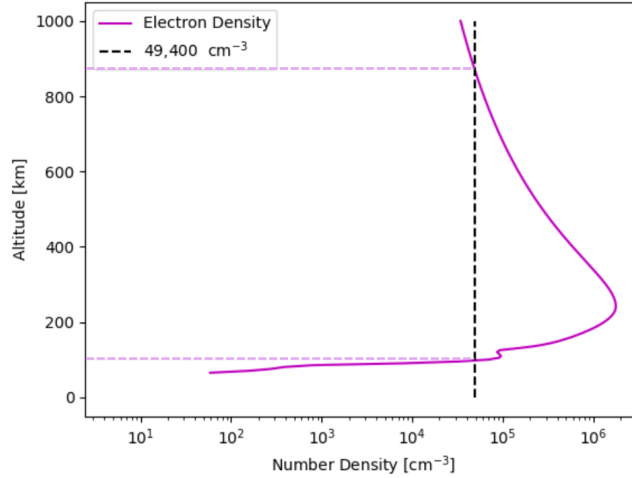
Frequencies above 12 MHz will be able to transmit through the ionosphere into space.

- (ii) The altitude where the maximum electron density is found in the IRI profile is 240 km. This is in the F-region.

(iii) First, we need to find the density that will reflect the 2 MHz radio wave:

$$n_e = \left(\frac{f_c}{9 \times 10^{-3}} \right)^2 = 49,400 \text{ cm}^{-3}$$

There are two heights that correspond to $49,400 \text{ cm}^{-3}$, as illustrated in the plot below. 870 km is the reflection altitude for transmission from space-to-ground. For ground-to-space transmission, the reflection would occur around 100 km.



6 Radio Blackout

Question During spacecraft re-entry, as demonstrated in Hidden Figures, the spacecraft temporarily loses the ability to communicate with the ground.

- (i) Describe in a few sentences why this happens.
- (ii) If the module normally communicates with the ground at 2282.5 MHz, what must the minimum electron density be in the plasma sheath region?

Answer (i) The air around the spacecraft ionizes due to the heat from the compression of the atmosphere by the spacecraft. When this ionized envelope has a high enough electron density, it reflects radio waves, just like the ionosphere.

- (ii) The minimum electron density to reflect a 2282.5 MHz radio wave:

$$n_e = \left(\frac{2282.5}{9 \times 10^{-3}} \right)^2 = 6.43 \times 10^{10} \text{ cm}^{-3}$$

7 GPS Delays

Question i) Calculate the path length delay through the ionosphere for 10 TECU, at L1 = 1575.42 MHz and at L2 = 1227.60 MHz. ii) Explain how you can use dual-frequency GPS to reduce the path length (and thus position) errors due to the ionosphere.

Answer (i) The path length delay is proportional to the ionospheric TEC and inversely proportional to the square of the GPS frequency:

$$\begin{aligned}\Delta l_{iono} &= -\frac{40.3 \frac{m^3}{s^2}}{f^2} TEC \\ \Delta l_{iono,L1} &= -\frac{40.3 \frac{m^3}{s^2} (10 \text{ TECU})}{(1575.42 \text{ MHz})^2} = -\frac{40.3 \frac{m^3}{s^2} (10 \times 10^{16} \text{ e-/m}^2)}{(1575.42 \times 10^6 \text{ Hz})^2} = \boxed{1.624 \text{ m}} \\ \Delta l_{iono,L2} &= -\frac{40.3 \frac{m^3}{s^2} (10 \text{ TECU})}{(1227.60 \text{ MHz})^2} = -\frac{40.3 \frac{m^3}{s^2} (10 \times 10^{16} \text{ e-/m}^2)}{(1227.60 \times 10^6 \text{ Hz})^2} = \boxed{2.674 \text{ m}}\end{aligned}$$

(ii) The phase difference between L1 and L2 can be used to estimate the TEC. To calculate the phase delay L_i , we subtract the path length delay from the integrated path length of the link, s_i :

$$L_i = s_i - \frac{40.3 \frac{m^3}{s^2}}{f_i^2} TEC$$

From the phase difference between the two signals, $L_1 - L_2$, we can solve for TEC. Note that s_1 should equal s_2 , as we are talking about two signals transmitted from the same GPS satellite to the same receiver:

$$\begin{aligned}L_1 - L_2 &= \left[s_1 - \frac{40.3 \frac{m^3}{s^2}}{f_1^2} TEC \right] - \left[s_2 - \frac{40.3 \frac{m^3}{s^2}}{f_2^2} TEC \right] \\ L_1 - L_2 &= s_1 - s_2 + \left(\frac{-1}{f_1^2} + \frac{1}{f_2^2} \right) \left(40.3 \frac{m^3}{s^2} TEC \right) \\ \frac{(L_1 - L_2)}{40.3 \frac{m^3}{s^2}} &= \left(\frac{f_1^2 - f_2^2}{f_1^2 f_2^2} \right) TEC \rightarrow \boxed{TEC = \frac{f_1^2 f_2^2 (L_1 - L_2)}{40.3 \frac{m^3}{s^2} (f_1^2 - f_2^2)}}\end{aligned}$$

As you can see, dual-frequency GPS uses the phase difference (really the path length difference) between the two signals to estimate TEC and ionospheric error.

8 D-region Signal Attenuation

Question Calculate the attenuation (in dB) of a 10 MHz signal propagating vertically through your IRI ionosphere from Problem 1. Use your MSIS atmosphere profile from the previous Module to determine the collision frequency, and assume $|B_0| = 50,000 \text{ nT}$.

Answer The attenuation, in dB, is found as follows:

$$\begin{aligned}\frac{dA}{dl} &= 4.6 \times 10^{-5} \frac{n_e \nu_{en}}{\nu_{en}^2 + (\omega \pm \Omega)^2} \\ \int dA &= 4.6 \times 10^{-5} \int \frac{n_e \nu_{en}}{\nu_{en}^2 + (\omega \pm \Omega)^2} dl \\ A[\text{dB}] &= 4.6 \times 10^{-5} \int \frac{n_e \nu_{en}}{\nu_{en}^2 + (\omega \pm \Omega)^2} dl\end{aligned}$$

We know $\omega = (2\pi \times 10 \times 10^6)$ rad/s, we have the n_e profile from the IRI model (convert to m^{-3}), and we need to determine ν_{en} , the electron collision frequency. We can estimate ν_{en} , which varies with altitude, as:

$$\nu_{en} = 1.2 \times 10^{11} \frac{N}{N_0}$$

where N is the neutral density at some altitude, and N_0 is the neutral density at the surface of the Earth. We can determine this from the MSIS profile provided for the last homework. Only including the altitudes that overlap in the MSIS and IRI profiles, we find attenuation values of about 0.3 dB & 67 dB (two answers because of the ' $\pm$ ' in the equation).

See the following code as a reference.

```
v_ne = np.array([1.2e11*n/n0 for n in total_rho]) # collision freq estimate
om = 2 * np.pi * 10e6 # 10 MHz wave, rad/s
q = 1.6e-19 # charge of an electron
B0 = 50000e-9 # 50,000 nT
m_e = 9.11e-31 # mass of an electron
om_c = 2 * np.pi * q*B0/m_e # rad/s
A_1 = 4.6e-5 * np.trapz((n_e*v_ne)/(v_ne**2 + (om + om_c)**2),x=alt)
A_2 = 4.6e-5 * np.trapz((n_e*v_ne)/(v_ne**2 + (om - om_c)**2),x=alt)
print(f'New Attenuation = {A_1:.3g} - {A_2:.3g} dB')
```

9 ICON Mission

Question The ICON Mission has four instruments: MIGHTI, IVM, EUV, and FUV (thank Berkeley for those creative acronyms). Do some research and pick your favorite instrument. In a couple of paragraphs, explain:

- What does this instrument do?
- What is the instrument SWAP (size, weight, and power)?
- In basic terms, how does it work?
- How does it contribute to the science goals of the ICON mission – in particular, what information about the ionosphere does it provide?

Answer Varies by selected instrument. Slack us if you have questions!